SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

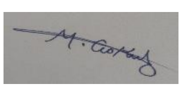
Annexure A

ARTICLE REVIEW

Code of Conduct:

I M.GOKUL 192565067__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article’s purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ARTICLE REVIEW

Article Title:
Decision Tree Application to Classification Problems with Boosting Algorithm

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Task 1: Article Summary

1(a) Full Citation of the Article

Authors: Long Zhao, Sanghyuk Lee, and Seon-Phil Jeong

Title: *Decision Tree Application to Classification Problems with Boosting Algorithm*

Journal: *Electronics*

Year: 2021

Volume and Issue: Volume 10, Issue 16

Article Number: 1903

DOI: 10.3390/electronics10161903

The study belongs to the area of **Machine Learning and Classification**, with a specific focus on **Decision Tree algorithms** and **Boosting techniques**. The application domain of the article is **personal credit evaluation**. The major machine learning problem addressed in the article is the classification of individuals based on their credit-related information.

Domain/Application Area

The domain of this research is **financial decision-making and credit risk classification**. Financial institutions often need to evaluate whether an individual can be classified into an appropriate credit category based on available information. Machine learning techniques can support this decision-making process by identifying patterns from historical data.

The article applies a **Classification and Regression Tree (CART)** model with a **boosting algorithm**. The purpose is to improve the classification performance and processing efficiency of a conventional decision-tree approach.

Specific ML Problem Addressed

The specific machine learning problem is a **classification problem**. The system takes relevant credit-related information as input and attempts to classify the data into appropriate classes. The authors investigate whether the addition of a boosting technique can improve the performance of a traditional CART decision-tree algorithm.

1(b) Article Summary and Key Objective

Summary

The main objective of the article is to investigate the effectiveness of combining a decision-tree algorithm with a boosting technique for solving classification problems in personal credit evaluation. Decision trees are widely used because they are easy to understand and can generate clear decision rules. However, a single decision tree may not always provide the best possible classification performance.

The authors therefore propose the use of the **Classification and Regression Tree (CART)** algorithm together with a **boosting algorithm**. Boosting is an ensemble learning technique that combines multiple learning processes to improve the overall predictive performance. The study compares the performance of the conventional CART algorithm with the boosted CART approach.

The research shows that the boosted CART model achieved an accuracy of **90.95%**, while the conventional CART approach achieved **90.31%**. The results indicate that boosting can provide a small improvement in classification accuracy and can also improve processing efficiency in the studied application.

The research is important because credit classification requires accurate and reliable decision-making. Incorrect classification can create financial risk for institutions and may also affect individuals. Therefore, improving the performance of machine learning models can be useful in financial decision-support systems.

Research Gap Identified

The major research gap addressed by the authors is that a traditional decision-tree model may have limitations in classification performance and efficiency when used independently. A single decision tree can be sensitive to the characteristics of the training data and may not always provide the most effective classification result.

The authors attempt to fill this gap by integrating a **boosting algorithm** with the **CART decision-tree algorithm**. Instead of depending only on a conventional decision tree, the proposed approach uses boosting to improve the learning process.

How the Authors Proposed to Fill the Gap

The authors proposed the following approach:

1. Use the CART algorithm as the basic decision-tree classifier.
2. Apply a boosting technique to improve the learning process.
3. Train and evaluate the boosted model.
4. Compare its performance with the conventional CART model.
5. Measure classification accuracy and analyze the effectiveness of the proposed method.

The results suggest that the boosted CART approach achieved better classification accuracy than the conventional decision tree in the studied experiment.

Task 2: Methodology Analysis

2(a) Machine Learning Technique Used

The main machine learning technique used in the article is the **Decision Tree**, specifically the **Classification and Regression Tree (CART)** algorithm combined with a **Boosting algorithm**.

CART Algorithm

CART is a supervised machine learning algorithm that can be used for classification and regression

problems. In this article, it is used for classification.
A decision tree works by dividing the dataset into smaller groups based on selected features. At every stage, the algorithm selects conditions that help separate the data into different classes.
The general structure of a decision tree contains:

- **Root Node** – The first and main decision point.
- **Internal Nodes** – Intermediate decision points.
- **Branches** – Connections representing decisions.
- **Leaf Nodes** – Final output or predicted class.

For example, a simplified credit classification rule can be represented as:
IF income is high AND credit history is good
THEN classify the person into a favorable credit category.
One major advantage of decision trees is **interpretability**. The generated decision rules can be easier for humans to understand compared with highly complex black-box models.

Boosting Algorithm
Boosting is an **ensemble machine learning technique**. Instead of relying on only one model, boosting improves the final prediction by combining multiple learning processes.
The basic idea is:

1. Train a model.
2. Identify the classification errors.
3. Give greater attention to difficult or incorrectly classified examples.
4. Build additional learning models.
5. Combine the learning results to produce an improved final model.

In this article, boosting is combined with the CART algorithm to improve the classification process.
Importance of Combining CART and Boosting
The combination provides two important benefits:

- The decision tree provides understandable decision rules.
- Boosting attempts to improve classification performance by improving the learning process.

Therefore, the proposed system combines the advantages of decision trees and ensemble learning.

Dataset Used
The article focuses on **personal credit evaluation data**. The dataset represents information relevant to a classification task in the financial domain.
The data preparation process is an important part of machine learning because the quality of the input data directly affects the quality of the prediction model.
The general data preparation process involves:
1. Data Collection
The relevant credit-related information is collected from available data sources.
2. Data Organization
The data is organized into a structured format so that the machine learning algorithm can process the information.
3. Feature Selection
Relevant features are used as input variables for the classification model.
4. Training
A part of the data is used to train the CART and boosted CART models.

5. Testing and Evaluation
The trained models are evaluated using classification performance measures.

2(b) Mapping the Technique to CO4 Topics
According to the course outcome, CO4 focuses on implementing machine learning techniques, including decision-tree algorithms, reinforcement learning, and models for classification and decision-making.
The selected article is strongly related to the following CO4 topics:

CO4 Topic	Connection to the Article
Decision Trees	The article uses the CART decision-tree algorithm.
Inductive Learning	The model learns general decision rules from available examples.
Classification	The system predicts or classifies data into categories.
Decision-Making	The model supports financial and credit-related decisions.
Machine Learning	The article applies supervised learning to real-world classification.
Ensemble Learning	Boosting is used to improve the decision-tree learning process.

Relationship with Inductive Learning
Inductive learning means learning general patterns from specific examples.
For example:

- Training data contains many examples.
- The model studies the relationship between input features and output classes.
- It creates general decision rules.
- These rules are then applied to new data.

In this article, the CART model learns classification patterns from the available data and applies the learned patterns to the classification problem.
Therefore, the study demonstrates an important principle of **inductive machine learning**.

Methodological Strength
One important methodological strength is the **comparison between the conventional CART model and the boosted CART model**.
This comparison allows the researchers to examine whether the boosting technique actually provides improvement. Instead of simply presenting one model, the authors compare:

- Traditional CART
- CART combined with Boosting

The results show that the boosted approach achieved a slightly higher accuracy.
Another important strength is that decision trees provide an understandable structure. This is useful in financial applications because decision-makers may want to understand why a particular classification result was generated.

Methodological Limitation
One limitation is that the reported improvement in accuracy is relatively small:

- Conventional CART: **90.31%**
- Boosted CART: **90.95%**

The difference is only **0.64 percentage points**.

Therefore, accuracy alone may not be sufficient to prove that the boosted method is significantly better in every practical situation.
The research could be strengthened by including additional evaluation measures such as:

- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix
- Cross-validation results

These metrics would provide a more complete understanding of the performance of the model.

Task 3: Results and Critical Evaluation
3(a) Key Results
The major experimental result reported in the article is the comparison between the conventional CART decision-tree model and the boosted CART approach.

Model	Classification Accuracy
Conventional CART	**90.31%**
CART with Boosting	**90.95%**

The boosted CART model performed better than the conventional CART model.
The improvement in accuracy was:
90.95% − 90.31% = 0.64%
Although the improvement is small, it indicates that the boosting technique improved the classification performance in the experiment.
The article also indicates that the boosting algorithm helped improve processing efficiency. Therefore, the proposed approach attempts to improve both the quality and effectiveness of classification.

Interpretation of the Results
The results suggest that combining boosting with a decision tree can improve machine learning performance.
A single decision tree learns patterns directly from the training data. However, boosting improves the learning process by focusing more on examples that are difficult to classify.
Therefore:
Conventional Decision Tree → Learns classification rules
Boosted Decision Tree → Learns and improves difficult classification cases
This can produce better overall classification performance.

3(b) Critical Evaluation of the Results
Are the Reported Metrics Realistic?
The reported accuracy values of approximately 90% are realistic for a machine learning classification problem, depending on the quality of the dataset and the difficulty of the classification task.
However, accuracy should not be considered the only evaluation measure.
For example, if one class is much more common than another class, a model may achieve high accuracy while performing poorly on the minority class.
Therefore, additional performance measures would provide stronger evidence.
Important Additional Metrics

Precision
Precision measures how many predicted positive cases are actually correct.
Recall
Recall measures how many actual positive cases were correctly identified.
F1-Score
The F1-Score combines precision and recall.
ROC-AUC
ROC-AUC measures the ability of the model to distinguish between classes.
These additional metrics would provide a more detailed evaluation than accuracy alone.

Potential Dataset Bias
One possible limitation is **dataset bias**.
The model performance may depend strongly on the characteristics of the dataset used for training and testing.
Possible issues include:

- Imbalanced classes.
- Limited data size.
- Missing values.
- Biased samples.
- Dataset-specific patterns.
- Lack of representation of different groups.

If the dataset is not sufficiently representative, the model may perform well during testing but may not achieve the same performance in real-world situations.

Potential Evaluation Bias
Another issue is the evaluation protocol.
If the model is tested only on data that is very similar to the training data, the reported accuracy may be optimistic.
The study could be improved by using:

- K-fold cross-validation.
- Multiple independent test sets.
- External validation.
- Different datasets.
- Statistical significance testing.

These experiments would help determine whether the improvement from 90.31% to 90.95% is consistent and meaningful.

Additional Experiments That Would Strengthen the Research
The following additional experiments could strengthen the findings:
1. Compare More Algorithms
The proposed model could be compared with:

- Random Forest
- Support Vector Machine
- Logistic Regression
- Naïve Bayes

- Gradient Boosting
- XGBoost

This would show whether boosted CART is competitive with other modern machine learning methods.

2. Perform Cross-Validation

K-fold cross-validation could provide more reliable results by evaluating the model using different partitions of the dataset.

3. Use Additional Metrics

The researchers could report:

- Precision
- Recall
- F1-Score
- ROC-AUC

4. Test on Additional Datasets

The model could be tested using multiple credit-related datasets.

5. Analyze Feature Importance

The importance of different input features could be analyzed to determine which factors have the greatest influence on classification.

3(c) Real-World Applicability

The technique proposed in the article has several real-world applications.

1. Banking

Banks can use machine learning to support decisions related to customer classification and credit evaluation.

2. Financial Institutions

Financial organizations can use classification models to identify patterns in customer-related data.

3. Credit Risk Analysis

Machine learning can help analyze large volumes of historical data and support risk-related decision-making.

4. Automated Decision Support Systems

Decision trees can be integrated into automated systems to provide faster preliminary classifications.

Challenges in Scaling the Solution to Industry

Although the technique is useful, several challenges exist.

1. Data Availability

Large, high-quality datasets are required to train reliable machine learning models.

2. Data Privacy

Financial data can contain sensitive information. Organizations must protect customer data and follow applicable privacy and security requirements.

3. Model Bias

A biased training dataset can produce unfair or unreliable classifications.

4. Data Changes Over Time

Real-world financial behavior can change. A model trained using old data may become less effective.

5. Model Monitoring

The system must be monitored regularly to ensure that its performance remains reliable.

6. Explainability

Financial institutions may need to explain automated decisions. Decision trees are useful because they are generally more interpretable than many complex machine learning models.

7. Computing Requirements

Larger datasets and more complex ensemble methods may require additional computational resources.

Task 4: Reflection and Learning Outcome

4(a) Three Key Insights

Insight 1: Decision Trees Learn Decision Rules from Data

The first major insight is that decision trees can automatically learn decision rules from training data. This is directly connected to **Inductive Learning**.

The algorithm studies individual examples and identifies general patterns. It then creates rules that can be applied to new examples.

Connection to CO4

Inductive Learning → Learning general patterns from specific examples

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Insight 2: Ensemble Techniques Can Improve Model Performance

The second insight is that combining machine learning techniques can improve the performance of a model.

The article demonstrates this by combining:

CART + Boosting

The boosted model achieved better accuracy than the conventional CART model.

This shows that a single machine learning model is not always the final solution. Ensemble methods can improve learning by combining multiple learning processes.

Connection to CO4

Decision Trees + Classification + Machine Learning Models

Insight 3: Machine Learning Performance Must Be Critically Evaluated

The third insight is that high accuracy alone does not always mean that a model is perfect.

A good machine learning evaluation should consider:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Dataset bias
- Evaluation procedure

This is important because a model may perform well on one dataset but perform differently in a real-world environment.

Connection to CO4

Classification and Decision-Making

The model must be evaluated carefully before it is used for important real-world decisions.

4(b) Proposed Novel Extension

Proposed Enhancement: Hybrid Credit Classification Model

If I were extending this research, I would propose a hybrid machine learning experiment that compares the boosted CART model with modern ensemble models.

Proposed Models

The experiment could compare:

1. Conventional CART
2. CART with Boosting
3. Random Forest
4. Gradient Boosting
5. XGBoost

Proposed Experiment

The following process could be followed:

Step 1: Collect a larger and more diverse credit dataset.
Step 2: Clean the dataset and handle missing values.
Step 3: Balance the classes if necessary.
Step 4: Divide the dataset into training and testing sets.
Step 5: Apply K-fold cross-validation.
Step 6: Train all models.
Step 7: Compare the results using multiple metrics.

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Processing Time

Feasibility of the Proposed Experiment

The proposed experiment is feasible because these machine learning algorithms are available in common software libraries such as Python machine learning frameworks.

The experiment can be performed using standard tools such as:

- Python
- Jupyter Notebook
- Pandas
- Scikit-learn

A suitable dataset can be divided into training and testing data, and multiple machine learning models can be trained and compared.

Expected Impact

The proposed extension could provide several benefits:

- Identify the most accurate classification model.
- Determine whether boosting provides a significant improvement.
- Improve the reliability of financial classification.
- Reduce the risk of selecting an unsuitable model.
- Improve understanding of different machine learning algorithms.
- Support better automated decision-making.

Conclusion

This article demonstrates an important application of **Decision Tree algorithms and Boosting techniques** in solving a classification problem. The study uses the CART algorithm together with a boosting technique for personal credit evaluation. The boosted CART model achieved **90.95% accuracy**, compared with **90.31% accuracy** for the conventional CART model.

The study is strongly related to the CO4 course outcome because it demonstrates **inductive learning, decision-tree classification, machine learning, and decision-making**. The research also shows that ensemble techniques can improve the performance of traditional machine learning algorithms.

From a critical perspective, the study would be stronger if it included additional evaluation metrics such as precision, recall, F1-score, and ROC-AUC, as well as more extensive validation experiments.

Overall, the article provides a useful example of how machine learning techniques can be applied to real-world classification problems and how decision-tree algorithms can be enhanced using boosting methods.

Reference

Zhao, L., Lee, S., & Jeong, S.-P. (2021). *Decision Tree Application to Classification Problems with Boosting Algorithm*. **Electronics, 10**(16), 1903. DOI: 10.3390/electronics10161903.